INTRODUCTION TO
ENGINEERING DESIGN-II

# Low-Cost Long-Range Outdoor Wi-Fi System

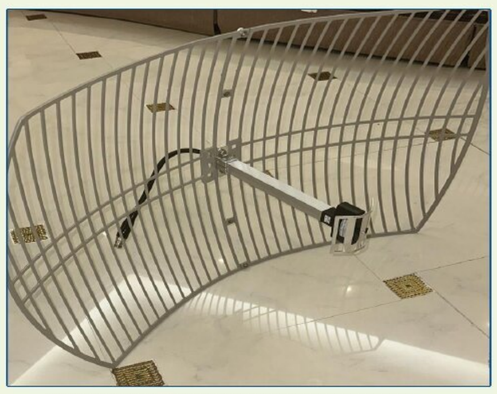

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## ABSTRACT


Our project aims to create and execute a Low-Cost Long-Range Wi-Fi System with an emphasis on amplifiers, repeaters, and antennas. The main objective is to develop a solution that is accessible and inexpensive for consumers with different levels of technical expertise. Our technology seeks to provide strong and long-lasting Wi-Fi coverage across wide outdoor regions by carefully planning antenna design, putting affordable repeater options into place, and sparingly integrating RF amplifiers. The goal of this project is to close the digital divide and provide affordable, long-range access to underserved populations.


## ARTIFACT



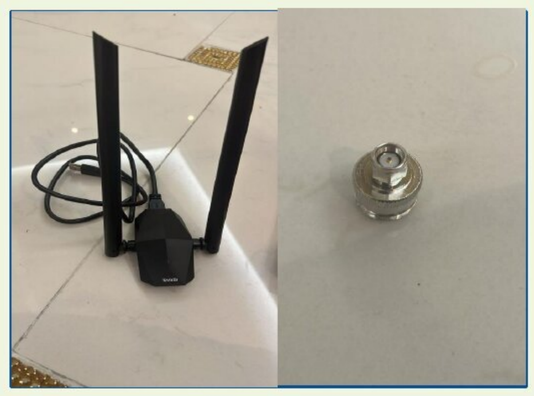
24dBi Square grid parabolic antenna



Wifi extender



Coaxial cable with N-type to SMA



USB adapter and SMA connectors

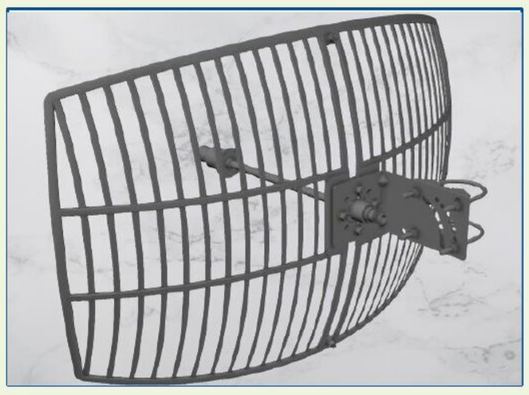


Two antennas connected to the extender

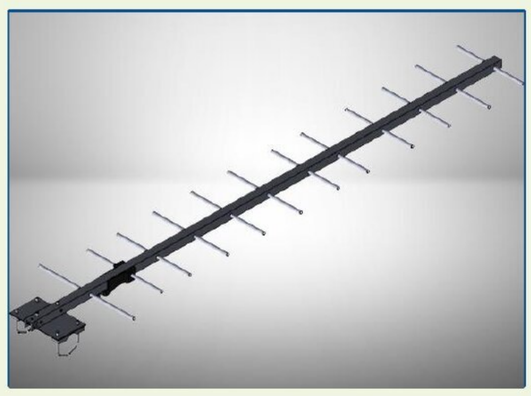


Extended Wifi signals reach the user

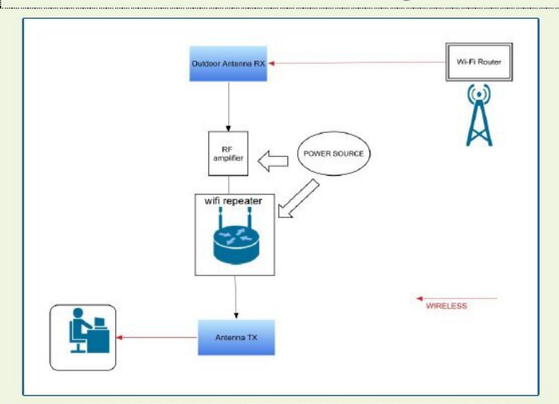
## PROJECT DEVELOPMENT



Square grid parabolic antenna solid works design.



YAGI antenna solid works design.




Block Diagram

Objectives

- Cost-effectiveness: Achieving economical operations.
- Long-distance effectiveness: Demonstrating strong performance over extended ranges.
- Deployment simplicity: Ensuring easy and straightforward implementation.

Constraints

- Adherence to financial constraints: The project must stay within the allocated budget.
- Completion within the semester timeframe: The task should be finished over the course of the semester.

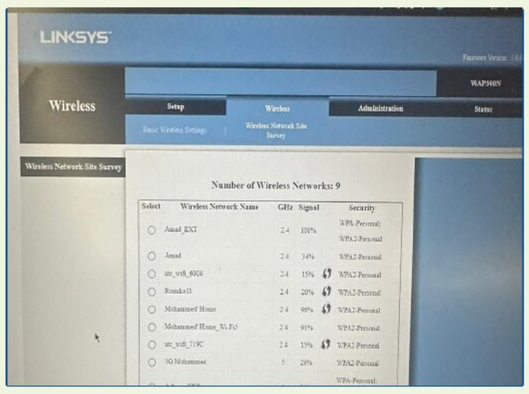
Objectives and Constraints

## FUTURE IMPROVEMENTS

1.Make the system smarter in handling signals.
2.Use components that save energy and space.
3.Include the latest technologies available.
4.Make it easier for users to set up and use, while also saving space.

## SIMULATION/MODELING APPROACH & RESULTS

Our project underwent crucial simulations, including speed tests and signal strength assessments. Before integrating our Low-Cost Long-Range Outdoor Wi-Fi System, the laptop received a certain number of signals with a specific download speed. Post-integration, we observed a substantial increase in the number of signals, multiplying significantly, and the download speed surged to an impressive 20 megabytes per second. These simulations underscore the tangible improvements achieved by our system in enhancing signal strength and download speeds over extended distances.




The number of wifi signals caught increase with strong signals after the connection

## ACKNOWLEDGMENT


We extend our sincere thanks to Dr. Ghaffer Kian for his guidance, King Abdulaziz University for providing lab access, "KSB Electrical Est" for supplying quality materials, and our dedicated team members for their time and contributions to our Low-Cost Long-Range Outdoor Wi-Fi System project


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